

Bioinformatics Best Practices

Home Folder Structure, Storage Limits & Blue Folder Usage

Advanced Molecular Detection Southeast Region

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In-Person Training

July 16–17, Athens, GA

- Two-day, hands-on workshop in Athens, GA
- Focus on genomic epidemiology tools and workflows
- Supports preparation for the 12-week training series
- Includes pipeline walkthroughs and small-group activities
- In-person or virtual training
- If you've not yet registered – please complete ASAP

[Registration](#)

Sanibel v2.0.1

- This is a small patch release with setup and documentation improvements, no pipeline logic changes:
 - Added environment.yaml file to facilitate conda installation using: `conda env create -f environment.yaml`
 - Nextflow is now pinned to version 25.10.4 in the sbatch file. Nextflow v26 introduced breaking changes and is not supported. Please make sure you are using a 25.x version.
 - Setup instructions were updated for kraken2 and bmgap2 for non-Florida BPHL users

Why This Matters?

- Keeps cluster/storage healthy
- Prevents Job Failures
- Speeds up workflows
- Helps track storage efficiently

Home Folder **Limit**

- **Hard** Limit: **40 GB**
- Recommended usage: ≤ 10 GB
- If you exceed 15 GB: schedule a 1-on-1 for cleanup
- Home Folder **is NOT** for ANY data or dependencies

Ex: Home Folder Format

- projects/ - scripts & config files
- docs/ - notes, metadata
- Symlinks (e.g. Blue) – Also add this so it can point to another directory.

```
[n.yengalareddy@login10 ~]$ df -h .
Filesystem                                Size  Used Avail Use% Mounted on
10.13.155.102:/home/n.yengalareddy      40G   3.1G   37G    8% /home/n.yengalareddy
[n.yengalareddy@login10 ~]$ ls
'~.'  bin  Blue  vscode_cli.tar.gz
```

Bad **vs** Good Example

Bad:

- Huge raw data stored
- FASTQC outputs, genome indexes
- Caches

Good:

- 1-5 GB Usage
- Organized; Not clustered with sensitive data

Blue Folder Usage

- All caches, dependencies and heavy operations must go here such as:
 - Conda caches
 - Genome Indexes
 - Singularity/Apptainer Images
 - Temporary storage for large data – once analyzed, raw should be moved to orange

Directory Decision

- Small Config or Script → Home Folder
- Large Dataset → Project or Data Directory in Blue Folder
- Software/cache → Blue Folder
- Temporary Work → Blue Folder

Additional **Best** Practices

- Compress rarely used files
- Delete any old logs
- Avoid copying large files → use symlinks
- Clear cache files
- **Don't** run pipelines from home folder

Demo

1. Check home-folder usage
2. Identify big files
3. Locate and clean unnecessary cache
4. Redirect cache to the Blue folder permanently
 - a. Redirect Nextflow cache
 - b. Redirect Singularity cache
 - c. Redirect BaseSpace CLI cache
5. Verification Step



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

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